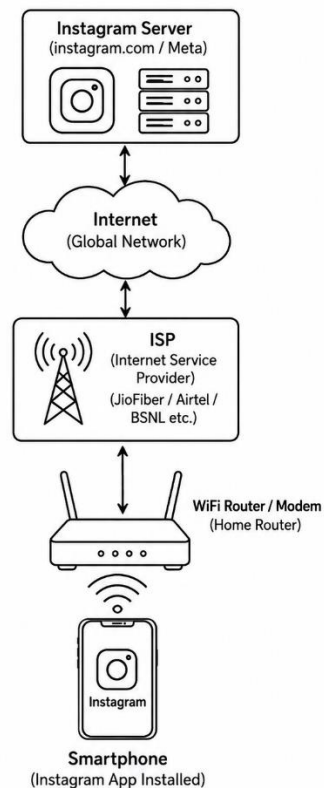


Introduction to Networks:-

Tasks:-

1. Draw a simple diagram (hand-drawn or digital) showing how your phone connects to Instagram, including your device, WiFi router, ISP, and Instagram server. Label each part and the connections.

Ans. :-



2. List 5 devices you use or see daily that are part of a network (for example: smartphone, smart TV, WiFi router, laptop, smartwatch). For each, write what type of network it connects to (LAN, WAN, PAN, etc.).

Ans. :-

Device	Type of Network	Explanation
Smartphone	WAN (Mobile Network) and LAN (WiFi)	Connects to the internet using mobile data (4G/5G) or to a home WiFi network.
Laptop	LAN (Local Area Network)	Connects to the home or office WiFi/Ethernet network to access the internet and share files.
WiFi Router	LAN and WAN	Creates the local network (LAN) and connects it to the internet through the ISP (WAN).
Smart TV	LAN	Connects to the home WiFi network to stream videos and access online apps.
Smartwatch	PAN (Personal Area Network)	Connects to a smartphone using Bluetooth, which is a Personal Area Network (PAN).

3. Explain the difference between the Internet and an intranet using an example from your college or a public WiFi you have used.

Ans.:-

Feature	Internet	Intranet
Definition	A global public network connecting computers worldwide.	A private network used within an organization.
Access	Open to anyone with an internet connection.	Restricted to authorized users only.
Users	General public.	Students, teachers, employees, or organization members.
Ownership	No single owner; managed by many organizations worldwide.	Owned and managed by a specific organization.
Security	Less secure because it is public.	More secure due to restricted access and authentication.
Purpose	Sharing information and communication worldwide.	Sharing internal information and resources within an organization.
Examples	Google, YouTube, Instagram, Gmail, Wikipedia.	College ERP, Student Portal, Library Management System, Employee Portal.

4. Pick any one app you use (like WhatsApp, Zomato, or Flipkart) and describe, in 4-5 lines, how data travels from your device to their server and back when you perform a search inside the app.

Hint: Mention the devices and networks involved, and what happens when you hit 'search'.

Ans.:-

Flipkart App :-

When I open the Flipkart app and search for a product, such as "wireless headphones," the app sends my search request from my smartphone through my WiFi router (or mobile network) to my Internet Service Provider (ISP). The ISP forwards the request over the Internet to Flipkart's servers, where the search is processed by checking the product database. The server then prepares a list of matching products, including their names, prices, ratings, images, and availability. This information is sent back through the Internet, ISP, and WiFi/mobile network to my smartphone. Finally, the Flipkart app displays the search results on the screen, allowing me to compare products and make a purchase.

Devices and Networks Involved:-

- Smartphone – Used to perform the search.
- WiFi Router or Mobile Network (4G/5G) – Connects the phone to the Internet.
- ISP (Internet Service Provider) – Provides Internet access.
- Internet – Transfers the data between the phone and the server.
- Flipkart Server – Processes the search request and sends back the results.

